Ex - 08
Data Visualization Using Tableau

AIM

To visualize data using Tableau by connecting to various data sources, creating bar charts, line charts, and pie charts, implementing calculated fields, and building interactive dashboards and stories using the Grains dataset.

PROBLEM STATEMENT

To analyze and visualize the Grains dataset using Tableau by connecting Excel, CSV, and SQL databases, creating basic charts, performing calculations, and developing interactive dashboards to present meaningful insights such as grain-wise sales, monthly trends, and category-wise distribution.

PROCEDURE

1. Open Tableau Desktop and explore the interface, including the Data Source page, worksheet, Rows shelf, Columns shelf, Marks card, and Show Me panel.
2. Connect to the Grains dataset (CSV) by selecting Text file under Connect, choosing the file, and loading the required data.
3. Connect to a SQL database by selecting the required database connector, entering the server details, providing authentication if needed, and selecting the required tables.
4. Check the imported data, verify the data types, rename fields if necessary, and remove unwanted columns or records to prepare the dataset for visualization.
5. Create a bar chart by dragging Grain to Rows and Sales to Columns, then select Bar Chart from the Show Me panel and sort the bars in descending order.
6. Create a line chart by dragging Order_Date (Month) to Columns and Sales to Rows, then select Line from the Marks card to display the monthly sales trend.
7. Create a pie chart by selecting Pie from the Marks card, dragging Category to Color, and dragging Sales to Angle to display category-wise sales proportions.
8. Create a calculated field by selecting Analysis > Create Calculated Field, entering a formula, such as [Quantity_kg] * [Price_per_kg], and clicking OK to generate a new calculated field.
9. Build a dashboard by selecting New Dashboard, dragging the required worksheets into the dashboard, and arranging charts, titles, legends, and filters to create an interactive layout.
10. Create a story by selecting New Story, adding worksheets or dashboards as story points, providing captions, and saving the workbook for further analysis and presentation.

IMPLEMENTATION

1. Calculated Field - Total Amount

```
[Quantity_kg] * [Price_per_kg]
```

2. Calculated Field - Sales Category

```
IF [Sales] >= 20000 THEN "High Sales"
ELSE "Low Sales"
END
```

Dashboard components:

• Bar chart - Grain-wise sales

• Line chart - Monthly sales trend

• Pie chart - Category-wise sales distribution (Cereal, Millet, Pulse)

• Stacked column chart - Region-wise sales by category

• Filter - Select a region, category or date range

• Story - Present visualizations in a sequence

OUTPUT

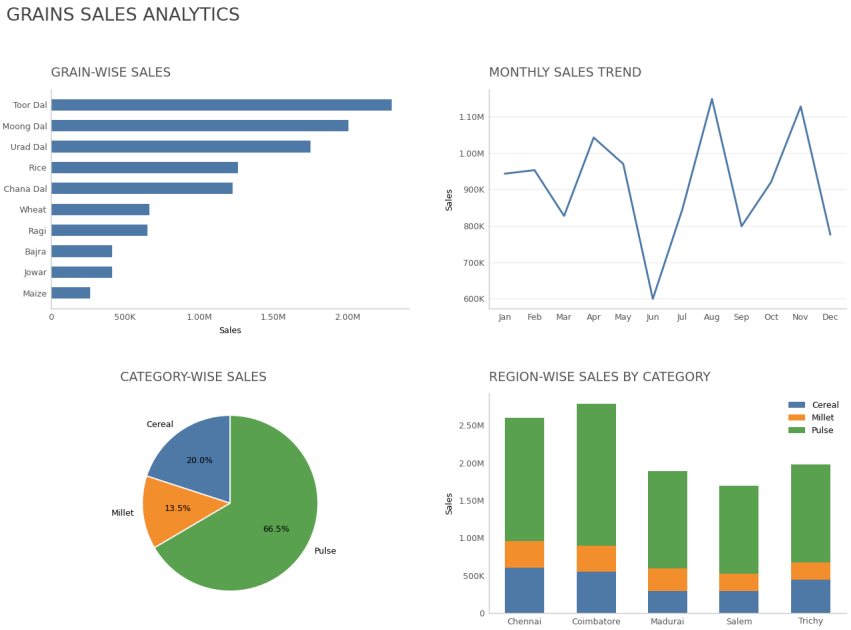


Fig: Grains Sales Analytics Dashboard (Tableau)

RESULT

Data visualization using Tableau was successfully studied and implemented on the Grains dataset. Data was connected from different sources, bar charts, line charts, and pie charts were created, calculated fields were implemented, and interactive dashboards and stories were developed to analyze and present meaningful insights.